

**Department of Pre Degree**  
**UTMSPACE**  
**Universiti Teknologi Malaysia**

*Test 2 (15%)*

*Time: 75 minutes*

**FSPM 0024 Calculus**

*Answer all questions.*

1. The derivative of a function  $g(x)$  is given as  $\frac{d(g(x))}{dx} = f(x) + x^2 - 1$ .

a) Express  $\int f(x) dx$  in term of  $g(x)$  and  $x$ .

(2 marks)

b) If  $g(1) = 3$  and  $g(2) = 7$ , find the value of  $\int_1^2 f(x) dx$ .

(3 marks)

2. Using appropriate substitution, evaluate  $\int \frac{2x+3}{(x+5)^3} dx$ .

(5 marks)

3. Use partial fractions to evaluate  $\int \frac{9}{x(3-2x)} dx$ .

(5 marks)

4. Evaluate  $\int x^2 \cos 3x dx$ .

(5 marks)

5. By the substitution  $x = 3 \sin \theta$ , evaluate  $\int_0^{\frac{3}{2}} \sqrt{9-x^2} dx$ .

(6 marks)

6. Find the area of the region bounded by the curve  $y = \sqrt{x}$ , the line  $x + y = 2$  and  $x$ -axis.

(4 marks)

$$1) \quad a) \int \frac{d}{dx} g(x) = \int f(x) + x^2 - 1$$

$$g(x) = \int f(x) + \int x^2 - \int 1$$

$$\int f(x) = g(x) - \frac{x^3}{3} + x$$

$$b) \quad g(1) = 3, g(2) = 7$$

$$\int_1^2 f(x) = g(x) - \int_1^2 x^2 + \int_1^2 1$$

$$= (g(2) - g(1)) - \left[ \frac{x^3}{3} \right]_1^2 + [x]_1^2$$

$$= 4 - \left( \frac{8}{3} - \frac{1}{3} \right) + 1$$

$$= \frac{8}{3}$$

$$2) \quad \int \frac{2x+3}{(x+5)^3} dx = \int \frac{2(u-5)+3}{u^3} du = \int \frac{2u-7}{u^3} du = \int 2u^{-2} du - \int 7u^{-3} du$$

$$u = x+5$$

$$x = u-5$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$= \frac{2u^{-1}}{-1} - \frac{7u^{-2}}{-2} + C$$

$$= -2u^{-1} + \frac{7u^{-2}}{2} + C$$

$$= -\frac{2}{u} + \frac{7}{2u^2} + C$$

$$= -\frac{2}{x+5} + \frac{7}{2(x+5)^2} + C$$

$$3) \quad \int \frac{9}{x(3-2x)} dx = \int \frac{3}{x} + \int \frac{6}{3-2x} = 3 \ln x + \int \frac{6^3}{u} x - \frac{du}{2} = 3 \ln x - 3 \ln(3-2x)$$

$$\frac{9}{x(3-2x)} = \frac{A}{x} + \frac{B}{3-2x} = \frac{3}{x} + \frac{6}{3-2x}$$

$$A(3-2x) + Bx = 9$$

$$3A - 2Ax + Bx = 9$$

$$3A = 9, A = 3$$

$$-6x + Bx = 0x$$

$$-6 + B = 0$$

$$B = 6$$

$$\text{let } u = 3-2x, \quad 2x = 3-u$$

$$\frac{du}{dx} = -2$$

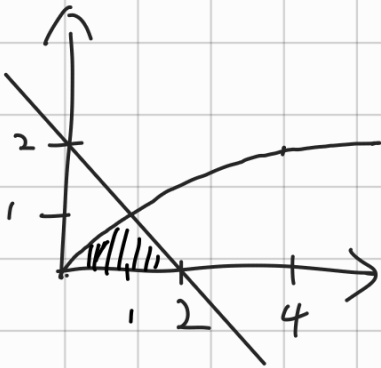
$$dx = -\frac{du}{2}$$

$$\begin{aligned}
 4) \int x^2 \cos 3x \, dx &= uv - \int v \frac{du}{dx} \, dx = x^2 \frac{\sin 3x}{3} - \int 2x \frac{\sin 3x}{3} \\
 u &= x^2 \quad \frac{du}{dx} = 2x \quad v = \frac{\sin 3x}{3} \quad \frac{dv}{dx} = \cos 3x \\
 &= x^2 \frac{\sin 3x}{3} - 2 \int x \frac{\sin 3x}{3} \\
 &= x^2 \frac{\sin 3x}{3} - 2 \left( -x \frac{\cos 3x}{9} + \int \frac{\cos 3x}{9} \, dx \right) \\
 &= \frac{x^2 \sin 3x}{3} + \frac{2x \cos 3x}{9} - \frac{\sin 3x}{27} + C
 \end{aligned}$$

$u=x \quad \frac{du}{dx} = \frac{\sin 3x}{3}$   
 $\frac{du}{dx} = 1 \quad v = -\frac{\cos 3x}{9}$

$$\begin{aligned}
 5) \int_0^{\frac{\pi}{2}} \sqrt{9-x^2} \, dx, \quad x &= 3 \sin \theta \quad \frac{dx}{d\theta} = 3 \cos \theta \\
 &= \int_0^{\frac{\pi}{2}} \sqrt{9-(3 \sin \theta)^2} \cdot 3 \cos \theta \, d\theta \quad dx = 3 \cos \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{2}} \sqrt{9-9 \sin^2 \theta} \, dx \cdot 3 \cos \theta \, d\theta \\
 &= \int_0^{\frac{\pi}{2}} \sqrt{9(1-\sin^2 \theta)} \cdot 3 \cos \theta \, d\theta \quad x = \frac{\pi}{2}, \theta = 0.5236 \\
 &= \int_0^{\frac{\pi}{2}} \sqrt{9 \cos^2 \theta} \cdot 3 \cos \theta \, d\theta \quad x = 0, \theta = 0 \\
 &= \int_0^{\frac{\pi}{2}} 9 \cos^2 \theta = \int_0^{\frac{\pi}{2}} \frac{9}{2} (1 + \cos 2\theta) \\
 &= \frac{9}{2} \left[ \theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\
 &= \frac{9}{2} (0.9567) = 4.304
 \end{aligned}$$

$$6) y = \sqrt{x}, \quad x+y=2, \quad x=y^2, \quad x=2-y$$



$$\begin{aligned}
 \text{Area} &= \int_0^1 2-y-y^2 \, dy \\
 &= \int_0^1 -y^2 - y + 2 \, dy \\
 &= \left[ -\frac{y^3}{3} - \frac{y^2}{2} + 2y \right]_0^1 \\
 &= \left( -\frac{1}{3} - \frac{1}{2} + 2 \right) - (0) \\
 &= 1.17 \text{ unit}^2
 \end{aligned}$$

$$\sqrt{x} = 2-x$$

$$x = 4 + x^2 - 4x$$

$$x^2 - 5x + 4 = 0$$

$$x_1 = 4, \quad x_2 = 1$$

| Trigonometric                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\cos^2 x + \sin^2 x = 1$<br>$1 + \tan^2 x = \sec^2 x$<br>$\cot^2 x + 1 = \operatorname{cosec}^2 x$<br>$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$<br>$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$<br>$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$<br>$\sin 2x = 2 \sin x \cos x$<br>$\cos 2x = \cos^2 x - \sin^2 x$<br>$= 2 \cos^2 x - 1$<br>$= 1 - 2 \sin^2 x$<br>$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$<br>$2 \sin x \cos y = \sin(x + y) + \sin(x - y)$<br>$2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$<br>$2 \cos x \cos y = \cos(x + y) + \cos(x - y)$ |
| Logarithm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $a^x = e^{x \ln a}$<br>$\log_a x = \frac{\log_b x}{\log_b a}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Differentiations                                                         | Integrations                                                              |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| $\frac{d}{dx}[k] = 0, k \text{ constant.}$                               | $\int k \, dx = kx + C, k \text{ constant.}$                              |
| $\frac{d}{dx}[x^n] = nx^{n-1}$                                           | $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, n \neq -1.$                    |
| $\frac{d}{dx}[e^x] = e^x.$                                               | $\int e^x \, dx = e^x + C.$                                               |
| $\frac{d}{dx}[\ln  x ] = \frac{1}{x}.$                                   | $\int \frac{dx}{x} = \ln  x  + C.$                                        |
| $\frac{d}{dx}[\cos x] = -\sin x.$                                        | $\int \sin x \, dx = -\cos x + C.$                                        |
| $\frac{d}{dx}[\sin x] = \cos x.$                                         | $\int \cos x \, dx = \sin x + C.$                                         |
| $\frac{d}{dx}[\tan x] = \sec^2 x.$                                       | $\int \sec^2 x \, dx = \tan x + C.$                                       |
| $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x.$                      | $\int \operatorname{cosec}^2 x \, dx = -\cot x + C.$                      |
| $\frac{d}{dx}[\sec x] = \sec x \tan x.$                                  | $\int \sec x \tan x \, dx = \sec x + C.$                                  |
| $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x.$ | $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C.$ |